INCH-POUND
MIL-M-38510/175B
AMENDMENT 4
23 APRIL 1999
SUPERSEDING
AMENDMENT 3
04 OCTOBER 1988

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS, POSITIVE LOGIC,
FLIP-FLOPS AND MONOSTABLE MULTIVIBRATOR,
MONOLITHIC SILICON,

Inactive for new design after 9 August 1996

This amendment forms a part of MIL-M-38510/175B, dated 30 April 1984,
and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2.3, add new case outline as follows:

"N F-5A (16-lead, .440" x .285" x .115"), flat package".

PAGE 2

1.4, add the following:

"The minimum value of external resistance, Rx, is 5 kΩ. The maximum value of external capacitance, Cx is 100 µF."

PAGE 4

* TABLE I, quiescent supply current ($I_{SS}$ or $I_{DD}$) , unit column: Delete "A" and substitute "µA".

PAGE 5

* TABLE I, input leakage current test, ($I_{IH}$), max limits column, device type 04, delete "45" and substitute "100". ($I_{IL}$), max limits column, device type 04, delete "-45" and substitute "-100".

PAGE 7

TABLE II, add "10,11" on line 8 for class B device. Delete line 12 in its entirety.

PAGE 8

FIGURE 1: Delete "cases E, F, and Z" and substitute "cases E, F, N, and Z" for device type 04.



FSC 5962

AMSC N/A
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PAGE 15

Delete truth table for device type 04 and substitute a new truth table as printed below.

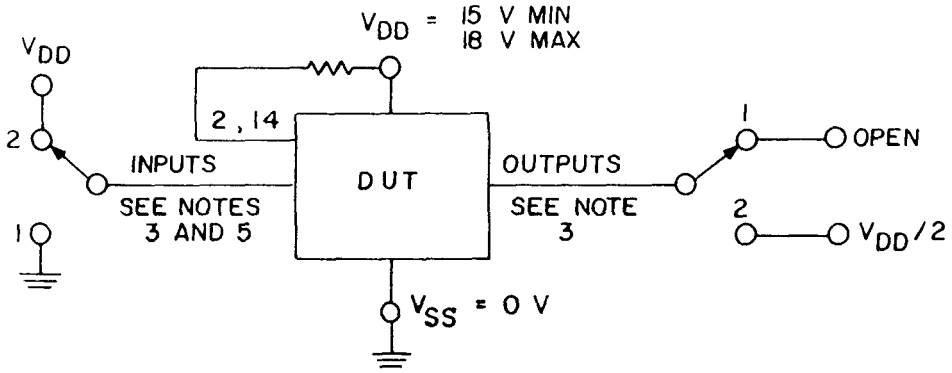
Device type 04

| Truth table | | | | | |
|---|---|---|---|---|---|
| Inputs | | | Outputs | | Timing |
| +Trigger | -Trigger | Reset | Q | $\overline{Q}$ | $R_X - C_X$ |
| L | H | L | L | H | H |
| _/‾ H | H | H | H | L | L |
| ‾_ L | H | H | N/C | N/C | L |
| L | H | H | L | H | H |
| L | ‾_ L | H | H | L | L |
| L | _/‾ H | H | N/C | N/C | L |

H = High level
L = Low level
N/C = No change

PAGE 16

FIGURE 4, static burn-in test circuit: Delete and substitute new static burn-in test circuit as printed below.




NOTES:
1. For static burn-in I, all inputs are connected to 0 volts, switch position 1.
2. For static burn-in II, all inputs are connected to $V_{DD}$, switch position 2.
3. Except for $V_{DD}$ and $V_{SS}$, terminals shall be connected through resistors whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20 % of its branded value due to use, heat or age.
4. Output may be in switch position 1 or 2.
5. Terminals 2 and 14 shall each be connected to VDD through separate 2 kΩ to 47 kΩ resistors. This requirement is only applicable to device type 04.



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PAGE 18

FIGURE 5, note 1: Delete first sentence and substitute: "Except for $V_{DD}$ and $V_{SS}$, terminals shall be connected through resistors whose value is 2 kΩ to 47 kΩ."

PAGE 23

FIGURE 6, notes: Delete "PRR = 877.193 kHz" and substitute "PRR 877 kHz", (two places).

PAGES 48 THROUGH 54

TABLE III: Delete "cases E, F, and Z" and substitute "cases E, F, N, and Z" for device type 04.

PAGE 50

* TABLE III, test number 66, subgroup 1, max limit column: Delete "6" and substitute "600".

* TABLE III, test numbers 67 through 72 subgroup 1, max limit column: Delete "1.0" and substitute "100"; subgroup 2, max limit column: Delete "45" and substitute "100".

* TABLE III, test number 73 subgroup 1, max limit column: Delete "-6 and substitute "-600".

PAGE 51

* TABLE III, test numbers 74 through 79 subgroup 1, max limit column: Delete "-1.0" and substitute "-100"; subgroup 2, max limit column: Delete "-45" and substitute "-100".



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TABLE III, delete and substitutes follows:

TABLE III. Group A inspection for device type 04.

| Symbol | MIL-STD-883 method | Cases E, F, Z | For terminal conditions and limits see 1/ | | | | | | | | | | | | | | | | | | Measured terminal | Test limits | | | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | Subgroup 1 $T_A$ = 25°C | | Subgroup 2 $T_A$ = 125°C | | Subgroup 3 $T_A$ = -55°C | | | | |
| | | | | | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | | | |
| | | Symbol | $CX_1$ | $RC_1$ | $R_1$ | $+TR_1$ | $-TR_1$ | $Q_1$ | $Q_1$ | $V_{SS}$ | $Q_2$ | $Q_2$ | $-TR_2$ | $+TR_2$ | $R_2$ | $RC_2$ | $CX_2$ | $V_{DD}$ | | | | | | | | | | | | |
| | | Test No. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | | | | | | | | | | | | |
| $V_{OH1}$<br>$V_{OH1}$ | 3006<br>3006 | 24<br>25 | | | GND<br>" | GND<br>" | 15.0 V<br>" | | | GND<br>" | | | 15.0 V<br>" | GND<br>" | GND<br>" | | | 15.0 V<br>" | Q1<br>Q2 | — | Q1<br>Q2 | — | 14.95<br>14.95 | | 14.95<br>14.95 | | 14.95<br>14.95 | | Vdc<br>" |
| $V_{OL1}$<br>$V_{OL1}$ | 3007<br>3007 | 26<br>27 | | | " | " | " | | | " | | | " | " | " | | | " | Q1<br>Q2 | | | | 0.05<br>0.05 | | 0.05<br>0.05 | | 0.05<br>0.05 | | " |
| $V_{OH1}$<br>$V_{OH1}$ | 3006<br>3006 | 28<br>29 | | 7/<br>" | 15.0 V<br>" | 15.0 V | " | | | " | | | " | 15.0 V<br>" | 15.0 V<br>" | 7/<br>" | | " | Q1<br>Q2 | | | | 14.95<br>14.95 | | 14.95<br>14.95 | | 14.95<br>14.95 | | " |
| $V_{OL1}$<br>" | 3007<br>3007 | 30<br>31 | | " | " | " | " | | | " | | | " | " | " | " | | " | Q1<br>Q2 | — | | | 0.05<br>0.05 | | 0.05<br>0.05 | | 0.05<br>0.05 | | " |
| $V_{OL1}$<br>$V_{OL2}$<br>$V_{LO3}$ | | 32<br>33<br>34 | | 10/<br>" | 10/<br>" | 10/<br>" | 10/<br>" | 10/<br>" | 10/<br>" | 10/<br>" | 10/<br>" | 10/<br>" | 10/<br>" | 10/<br>" | 10/<br>" | 10/<br>" | 10/<br>" | 5.0 V<br>10.0 V<br>15.0 V | All outputs | | | | 10/<br>" | | 10/<br>" | | 10/<br>" | | " |
| $V_{IH1}$<br>$V_{IH2}$<br>$V_{IH3}$ | | 35<br>36<br>37 | " | " | " | " | " | " | " | " | " | " | " | " | " | " | " | 5.O V<br>10.0 V<br>15.0 V | " | | | | " | | " | | " | | " |
| $I_{OL1}$ | | 38<br>39<br>40<br>41<br>42<br>43 | | 5.0 V<br>5.0 V<br>8/<br>"<br>" | GND<br>GND<br>GND<br>5.0 V<br>5.0 V<br>" | GND<br>"<br>"<br>5.0 V<br>5.0 V | GND<br>GND<br>5.0 V<br>"<br>" | 0.4 V | 0.4 V | " | 0.4 V | 0.4 V | GND<br>GND<br>5.0 V<br>"<br>" | GND<br>"<br>"<br>5.0 V<br>5.0 V | GND<br>GND<br>GND<br>5.0 V<br>5.0 V | 5.0 V<br>5.0 V<br>8/<br>"<br>" | " | 5.0 V<br>"<br>"<br>"<br>"<br>" | Q1<br>Q2<br>None<br>None<br>Q1<br>Q2 | —<br>—<br>—<br>—<br>—<br>— | | | 0.36<br>"<br>"<br>"<br>"<br>" | | 0.64<br>"<br>"<br>"<br>"<br>" | | | | mA<br>"<br>"<br>"<br>"<br>" |
| $I_{OH1}$ | | 44<br>45<br>46<br>47<br>48 | | 4.6 V<br>5.0 V<br>5.0 V<br>5.0 V<br>8/ | GND<br>"<br>"<br>"<br>" | GND<br>"<br>"<br>"<br>" | GND<br>"<br>"<br>"<br>5.0 V | 4.6 V | 4.6 V | " | 4.6 V | 4.6 V | GND<br>"<br>"<br>GND<br>5.0 V | GND<br>"<br>"<br>"<br>" | GND<br>"<br>"<br>"<br>" | 5.0 V<br>5.0 V<br>5.0 V<br>4.6 V<br>8/ | " | " | $RC_1$<br>$Q_1$<br>$Q_2$<br>$RC_2$<br>None | —<br>—<br>—<br>—<br>— | | | -0.36<br>"<br>"<br>"<br>"<br>" | | -0.64<br>"<br>"<br>"<br>"<br>" | | | | "<br>"<br>"<br>"<br>" |



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TABLE III, delete and substitutes follows:

TABLE III. Group A inspection for device type 04 – Continued.

| Symbol | MIL-STD-883 method | Cases E, F, Z Symbol | Test No. | For terminal conditions and limits see 1/ CX1 | RC1 | R1 | -TR1 | +TR1 | Q1 | Q1 | VSS | Q2 | -TR2 | +TR2 | R2 | RC2 | CX2 | VDD | Measured terminal | Test limits: Subgroup 1 TA = 25°C Min | Subgroup 1 TA = 25°C Max | Subgroup 2 TA = 125°C Min | Subgroup 2 TA = 125°C Max | Subgroup 3 TA = -55°C Min | Subgroup 3 TA = -55°C Max | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $I_{IL2}$ | 3009 | | 74 | | | GND | 18.0 V | 18.0 V | | | GND | | 18.0 V | 18.0 V | | | | | 18.0 V | | -100 | | -100 | | nA |
| | | | 75 | | | 18.0 V | GND | 18.0 V | | | " | | 18.0 V | 18.0 V | | | | | " | | " | | " | | " |
| | | | 76 | | | " | 18.0 V | " | | | " | | GND | " | | | | | " | | " | | " | | " |
| | | | 77 | | | " | GND | " | | | " | | 18.0 V | GND | | | | | " | | " | | " | | " |
| | | | 78 | | | " | 18.0 V | " | | | " | | 18.0 V | 18.0 V | | | | | " | | " | | " | | " |
| | | | 79 | | | " | 18.0 V | " | | | " | | 18.0 V | " | | | | | " | | " | | " | | " |
| | | | | | | | | | | | | | | | | | | | | Subgroup 4 TA = 25°C | | | | | |
| $C_I$ | 3012 | | 80 | | | 6/ | 6/ | 6/ | | | GND | | 6/ | 6/ | 6/ | | | | $R_1$ | Min | Max | | | | | pF |
| | | | 81 | | | " | " | " | | | " | | " | " | " | | | | $+TR_1$ | | | | | | | |
| | | | 82 | | | " | " | " | | | " | | " | " | " | | | | $-TR_1$ | | | | | | | |
| | | | 83 | | | " | " | " | | | " | | " | " | " | | | | $-TR_2$ | | | | | | | |
| | | | 84 | | | " | " | " | | | " | | " | " | " | | | | $+TR_2$ | | | | | | | |
| | | | 85 | | | " | " | " | | | " | | " | " | " | | | | $R_2$ | | | | | | | |
| | | | | | | | | | | | | | | | | | | | | Subgroup 7 | | Subgroup 8 | | | | |
| | | | | | | | | | | | | | | | | | | | | $T_A$ = 25°C | | $T_A$ = 125°C | | $T_A$ = -55°C | | |
| | | | | | | | | | | | | | | | | | | | | Min | Max | Min | Max | Min | Max | |
| Truth table tests | 3014 | | 86 | L | L | L | H | L | L | L | H | L | L | L | L | L | L | L | GND | NOTES | All outputs | See notes 4, 5, and 10 | | | | |
| | | | 87 | L | L | H | H | H | L | L | H | H | L | L | L | L | L | L | " | " | " | | | | | |
| | | | 88 | L | L | H | H | H | L | L | H | H | L | L | L | H | L | L | " | " | " | | | | | |
| | | | 89 | L | L | H | H | H | L | L | H | H | L | L | L | H | L | L | " | " | " | | | | | |
| | | | 90 | L | H | H | H | H | L | L | H | H | L | L | L | H | L | L | " | " | " | | | | | |
| | | | 91 | L | L | L | H | L | L | L | H | L | L | L | L | L | L | L | " | " | " | | | | | |
| | | | 92 | L | L | L | H | L | L | L | H | L | L | L | L | L | L | L | " | " | " | | | | | |
| | | | 93 | L | L | L | H | L | L | L | H | L | L | L | L | L | L | L | " | " | " | | | | | |
| | | | 94 | L | H | L | H | L | L | L | H | L | L | L | L | L | L | L | " | " | " | | | | | |
| | | | 95 | L | L | H | H | H | L | L | H | H | L | L | L | H | L | L | " | " | " | | | | | |
| | | | 96 | L | L | H | H | H | L | L | H | H | L | L | L | H | L | L | " | " | " | | | | | |
| | | | 97 | L | L | H | H | H | L | L | H | H | L | L | L | H | L | L | " | " | " | | | | | |
| | | | 98 | L | L | H | H | H | L | L | H | H | L | L | L | H | L | L | " | " | " | | | | | |
| | | | 99 | L | L | H | H | H | L | L | H | H | L | L | L | H | L | L | " | " | " | | | | | |
| | | | 100 | L | L | L | H | L | L | L | H | L | L | L | L | L | L | L | " | " | " | | | | | |
| | | | 101 | L | L | L | H | L | L | L | H | L | L | L | L | L | L | L | " | " | " | | | | | |
| | | | 102 | L | L | L | H | L | L | L | H | L | L | L | L | L | L | L | " | " | " | | | | | |
| | | | 103 | L | H | L | H | L | L | L | H | L | L | L | L | L | L | L | " | " | " | | | | | |
| | | | 104 | L | L | H | H | H | L | L | H | H | L | L | L | H | L | L | " | " | " | | | | | |
| | | | 105 | L | L | H | H | H | L | L | H | H | L | L | L | H | L | L | " | " | " | | | | | |
| | | | 106 | L | L | H | H | H | L | L | H | H | L | L | L | H | L | L | " | " | " | | | | | |
| | | | 107 | L | L | H | H | H | L | L | H | H | L | L | L | H | L | L | " | " | " | | | | | |
| | | | 108 | L | L | H | H | H | L | L | H | H | L | L | L | H | L | L | " | " | " | | | | | |
| | | | 109 | L | L | H | H | H | L | L | H | H | L | L | L | H | L | L | " | " | " | | | | | |



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PAGES 54 AND 55

Notes: Delete notes 5, 7, 8, and 10 and substitute the following:

"5/ The truth table test shall be performed at $V_{DD} \leq 5$ V and $\geq 18$ V. See note 10 which contains complete tabulation of input/output voltage levels.

"7/ For tests 28 through 31, terminals 2 and 14 are each connected through separate 22 kΩ ±10 % resistors to $V_{SS}$.

"8/ Terminals 2 and 14 are each connected through separate 22 kΩ ±10 % resistors to $V_{SS}$.

"10/ The following terminal conditions shall apply. The tests shall be performed using the logical timing sequence of test numbers 86 through 109. The parenthetical terminal conditions are used for the 'truth nonparenthetical terminal conditions are used for the '$V_{IH1}$, 2, 3/$V_{IL1}$, 2, 3' tests. The $RX_1$, $RX_2$ terminal conditions are for both of these tests.

Input/output conditions.

| $V_{DD}$ | | | Output levels | | | |
|---|---|---|---|---|---|---|
| | H | L | H | L | H | L |
| (≤ 5.0 V)<br>5.0 V | Max | Min<br>1.5 V<br>($V_{SS}$) | Min<br>($V_{DD}$ – 0.5 V)<br>4.5 V | Max<br>($V_{SS}$ + 0.5 V)<br>0.5 V | $V_{DD}$<br>through<br>22 k Ω<br>±10% | $V_{SS}$<br>through<br>22 k Ω<br>±10% |
| | ($V_{DD}$)<br>3.5 V | | | | | |
| 10.0 V | | 3.0 V | | | | |
| | 7.0 V | | 9.0 V | 1.0 V | | |
| 15.0 V<br><br>(≥ 18.0 V) | | 4.0 V<br>($V_{SS}$) | | | | |
| | ($V_{DD}$)<br>11.0 V | | 13.5 V<br>($V_{DD}$ – 0.5 V) | 1.5 V<br>($V_{SS}$ + 0.5 V) | | |



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Following 12/, add note 13/ as printed below.

"13/ The following timing sequence and input/output conditions shall apply:

| Input/output conditions | | | | | |
|---|---|---|---|---|---|
| Test | $V_{DD}$ | Input levels | | Output levels | |
| | | 1 | 0 | 1 | 0 |
| $V_{IL1}$<br>$V_{IH1}$ | 5.0 V | 3.5 V | 1.5 V | Min<br>4.5 V | Max<br>0.5 V |
| $V_{IL2}$<br>$V_{IH2}$ | 10.0 V | 7.0 V | 3.0 V | 9.0 V | 1.0 V |
| $V_{IL3}$<br>$V_{IH3}$ | 10.0 V | 11.0 V | 4.0 V | 13.5 V | 1.5 V |

| | Time Slot | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\overline{C}$ | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 |
| I | $D_1$ | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 0 |
| N | $D_2$ | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 0 |
| P | $D_3$ | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 0 |
| U | CLK | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 0 |
| T | $D_4$ | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 0 |
| S | $D_5$ | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 0 |
| | $D_6$ | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 0 |
| O | $Q_1$ | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 |
| U | $Q_2$ | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 |
| T | $Q_3$ | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 |
| P | $Q_4$ | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 |
| U | $Q_5$ | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 |
| T | $Q_6$ | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 |
| S | | | | | | | | | | | | | | | |

Test shall be run in sequence."



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4.4.1, add new paragraph as follows:

"e. Subgroups 9 and 11 shall be measured only for initial qualification and after process or design changes which may affect dynamic performance."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Navy - AS, CG, MC, OS, SH
Army - AR, MI, SM
Air Force - 19, 99

(Project 5962-1846)



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